VIRTUAL FENCING & NAVIGATION SYSTEM FOR AUTONOMOUS ROVER

PROJECT NAME:	Virtual fencing & navigation system for autonomous mobile rover
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WHY THIS PROJECT ?

- Physical fencing is costly, inflexible, and difficult to modify for changing environments. This project provides a software-based virtual fencing solution that allows easy boundary updates without physical infrastructure
- It reduces human supervision by enabling autonomous navigation within safe, predefined area.

INTRODUCTION

- This project presents an autonomous mobile rover that navigates within virtual boundaries using positioning and sensor data.
- The system ensures safe movement, obstacle avoidance, and controlled operation without the need for physical barriers.

KEY FEATURES

- Virtual fencing (geo-fencing) restricts the rover's movement within defined zones.
- Autonomous navigation with real-time obstacle detection ensures safe and efficient operation.

ARCHITECTURE

- The rover consists of sensors (GPS, ultrasonic/IR), a microcontroller, motor drivers, and a power system.
- Navigation algorithms process boundary data and sensor inputs to control rover movement and decision-making.

RESULTS

- The rover successfully operated within the virtual fence without crossing restricted areas.
- Accurate navigation and stable movement were achieved in real-time test environments

FUTURE SCOPE

1. **Warehouse automation**
2. **Smart surveillance systems**
3. **It can be applied in agriculture, security patrols**